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<title>The Human Insights Engine: how virtual personas are built</title>
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*,*::before,*::after{box-sizing:inherit}
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a{color:var(--rasp)}
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.wide{max-width:1180px;margin:0 auto;padding:0 24px}
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.hero svg.orb{position:absolute;right:-120px;top:-140px;width:520px;height:520px;opacity:.35}
.hero .wrap{padding-top:56px;padding-bottom:64px;position:relative}
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.brand svg{width:52px;height:52px}
.brand i{width:2px;height:42px;background:rgba(248,234,241,.7)}
.brand span{font:700 19px/1.05 'Funnel Display',sans-serif;color:#fff}

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22px;font-weight:700}
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.tag{margin:28px 0 0;font:600 15px 'Plus Jakarta Sans',sans-serif;color:#9FE0F4}
/* body */
main{padding:56px 0 24px}
h2{font-size:34px;line-height:1.1;margin:64px 0
18px;font-weight:700;display:flex;gap:14px;align-items:baseline}
h2 .n{font:700 18px 'Plus Jakarta
Sans',sans-serif;color:var(--cyanT);font-variant-numeric:tabular-nums}
h3{font-size:22px;line-height:1.2;margin:32px 0 10px;font-weight:700}
p{margin:0 0 18px}
.lead{font-size:20px;color:var(--ink)}
strong{color:var(--ink);font-weight:700}
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Display',sans-serif;color:var(--rasp);letter-spacing:-.02em}
.stat
span{display:block;margin-top:10px;font-weight:700;color:var(--ink);font-size:15px;line-height:1.
3}
.stat small{display:block;margin-top:6px;font-size:14px;line-height:1.45;color:var(--mut)}
.chain{display:flex;flex-wrap:wrap;gap:10px 28px;align-items:center;margin:22px 0
8px;padding:0;list-style:none}
.chain li{position:relative;font:600 15px 'Plus Jakarta
Sans',sans-serif;color:var(--ink);background:var(--card);border:1px solid
var(--line);border-radius:999px;padding:7px 14px;white-space:nowrap}
.chain
li:not(:last-child)::after{content:">";position:absolute;right:-19px;top:50%;transform:translateY(-54
%);color:var(--pink);font-weight:700;font-size:18px}
.chain.tags{gap:10px}.chain.tags li::after{content:none}
.chain.tags li{background:var(--tint);border-color:transparent}
.grid4{display:grid;grid-template-columns:repeat(4,1fr);gap:14px;margin:20px 0}
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.grid3{display:grid;grid-template-columns:repeat(3,1fr);gap:14px;margin:20px 0}
.card{background:var(--card);border:1px solid var(--line);border-radius:16px;padding:20px}
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.card p{font-size:15px;line-height:1.55;margin:0 0 10px}
.card .ex{font-size:14px;line-height:1.5;color:var(--rasp);font-weight:600;margin:0}
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.card.tint2{background:var(--tint2);border-color:transparent}
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Sans',sans-serif;color:var(--cyanT);display:block;margin-bottom:6px}

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0;color:var(--ink);font-weight:600;font-size:17px;line-height:1.55}
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1fr;gap:10px 28px;counter-reset:r}
ol.rules li{counter-increment:r;font-size:16px;line-height:1.5;padding-left:40px;position:relative}
ol.rules
li::before{content:counter(r);position:absolute;left:0;top:1px;width:26px;height:26px;border-radiu
s:50%;background:var(--rasp);color:#fff;font:700 13px/26px 'Plus Jakarta
Sans',sans-serif;text-align:center}
ol.rules b{color:var(--ink);display:block}
.flow{font-size:16px;line-height:1.7;color:var(--ink);background:var(--card);border:1px solid
var(--line);border-radius:16px;padding:18px 22px;margin:18px 0}
.flow em{font-style:normal;color:var(--pink);font-weight:700;padding:0 4px}
.weights{display:flex;height:44px;border-radius:12px;overflow:hidden;margin:22px 0 10px}
.weights div{display:flex;align-items:center;justify-content:center;color:#fff;font:700 15px 'Plus
Jakarta Sans',sans-serif}
.wlegend{display:grid;grid-template-columns:repeat(3,1fr);gap:8px 20px;margin:0 0
18px;padding:0;list-style:none;font-size:15px;line-height:1.45}
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.wlegend i{flex:0 0 12px;height:12px;border-radius:3px;margin-top:5px}
.wlegend b{color:var(--ink)}
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Display',sans-serif;color:var(--rasp);display:block;margin:4px 0 8px}
.cols{display:grid;grid-template-columns:1fr 1fr;gap:14px;margin:20px 0}
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/* embedded animation */
figure.hie{margin:34px auto 44px;max-width:1180px;padding:0 24px;outline:none}
figure.hie:focus-visible .frame{box-shadow:0 0 0 3px var(--pink)}
.frame{position:relative;width:100%;aspect-ratio:16/9;overflow:hidden;border-radius:18px;box-sh
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.stage{position:absolute;left:0;top:0;width:1920px;height:1080px;transform-origin:0
0;overflow:hidden;color:#4A0F2E;font-family:'Plus Jakarta Sans',sans-serif;
background:radial-gradient(900px 620px at 58% 64%,rgba(255,255,255,.7),transparent
70%),linear-gradient(118deg,#F7E6EE 0%,#F1EAF2 42%,#E6F1F5 100%)}
.stage::before{content:"";position:absolute;inset:0;background-image:radial-gradient(rgba(74,15,
46,.07) 1.2px,transparent 1.5px);background-size:28px
28px;-webkit-mask-image:radial-gradient(ellipse 70% 70% at 58% 62%,#000 10%,transparent
78%);mask-image:radial-gradient(ellipse 70% 70% at 58% 62%,#000 10%,transparent 78%)}
.stage canvas{position:absolute;left:0;top:0;width:1920px;height:1080px;z-index:1}
.stage header{position:absolute;left:96px;top:74px;width:1340px;z-index:2}

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.stage h5{font:700 62px/1.04 'Funnel Display','Plus Jakarta
Sans',sans-serif;letter-spacing:-.02em;margin:0;color:#4A0F2E;white-space:nowrap}
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0;font-weight:500;white-space:nowrap}
.stage
.logo{position:absolute;right:96px;top:78px;display:flex;align-items:center;gap:16px;z-index:2}
.stage .logo svg{width:72px;height:72px}.stage .logo
i{width:2px;height:58px;background:#4A0F2E;opacity:.8}
.stage .logo span{font:700 25px/1.05 'Funnel Display',sans-serif;color:#4A0F2E}
.stage .rail{position:absolute;left:96px;right:96px;top:238px;display:flex;gap:14px;z-index:3}
.stage .rail
button{flex:1;min-width:0;background:none;border:0;padding:0;text-align:left;color:#6E5664;font:
600 21px 'Plus Jakarta Sans',sans-serif;cursor:pointer;white-space:nowrap}
.stage .rail button:hover,.stage .rail button.on{color:#4A0F2E}
.stage .rail
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ottom:12px}
.stage .rail .bar b{display:block;height:100%;width:0;background:#EC6A9E}
.stage .rail button.on .bar b{background:#C2185B}
.stage .rail .n{opacity:.55;margin-right:10px}
.hbar{display:flex;align-items:center;gap:6px;margin-top:14px}
.hbar button{flex:0 0 auto;height:40px;min-width:40px;border-radius:20px;border:1px solid
var(--line);background:var(--card);color:var(--ink);display:grid;place-items:center;cursor:pointer;p
adding:0}
.hbar button:hover{border-color:var(--rasp)}
.hbar button:focus-visible{outline:2px solid var(--pink);outline-offset:2px}
.hbar .play{display:flex;gap:8px;padding:0 16px 0 12px;font:600 14px 'Plus Jakarta
Sans',sans-serif}
.hbar svg{width:18px;height:18px}
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10px;font-size:15px;line-height:1.45;color:var(--body);min-height:44px;display:flex;align-items:ce
nter}
figcaption{font-size:14px;color:var(--mut);margin-top:8px}
figure.hie:fullscreen{max-width:none;margin:0;padding:24px;background:#EFE5EB;display:flex;f
lex-direction:column;justify-content:center}
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figure.hie:fullscreen .hbar{width:min(100%,calc((100vh - 120px) * 16 / 9));margin:14px auto 0}
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@media
(max-width:560px){body{font-size:17px}.stats,.grid4,.grid2,.studies,.cols,ol.rules{grid-template-c
olumns:1fr}.wlegend{grid-template-columns:1fr}h2{font-size:28px}.hbar .cap{display:none}}
</style>

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  fill="none" stroke="#EC6A9E" stroke-width=".6"/><circle cx="50" cy="50" r="27" fill="none"
  stroke="#5FCDEA" stroke-width=".6"/><circle cx="82" cy="20" r="2.2" fill="#E0457F"/><circle
  cx="74" cy="41" r="1.8" fill="#5FCDEA"/></svg>
  <div class="wrap">
    <div class="brand"><svg viewBox="0 0 100 100" aria-hidden="true"><circle cx="50" cy="50"
    r="44" fill="none" stroke="#EC6A9E" stroke-width="2.4"/><circle cx="50" cy="50" r="27"
    fill="none" stroke="#5FCDEA" stroke-width="2.4"/><circle cx="50" cy="50" r="12"
    fill="#5FCDEA"/><circle cx="82" cy="20" r="5.5" fill="#E0457F"/><circle cx="8" cy="42" r="5"
    fill="#F29BBE"/><circle cx="74" cy="41" r="4.6" fill="#5FCDEA"/><circle cx="33" cy="30" r="3"
    fill="#fff" stroke="#EC6A9E" stroke-width="1"/><circle cx="36" cy="72" r="3.2"
    fill="#F29BBE"/><circle cx="76" cy="87" r="5" fill="#F29BBE"/></svg><i></i><span>The
    Human<br>Insights Engine</span></div>
    <h1>From real-world evidence to dynamic virtual worlds</h1>
    <p class="stand">How the Human Insights Engine turns evidence into bounded,
    context-aware, persistent in memory, and auditable virtual personas, and how it checks that
    what they tell us holds up.</p>
    <p class="tag">Real world evidence, simulated feedback, informed decisions.</p>
  </div>
</header>

<main>
<div class="wrap">
<p class="lead">The Human Insights Engine (HIE) is an evidence-informed and AI-powered
simulation tool that helps development organisations explore potential responses to a proposed
policy, program, product, service or message before resources are committed. It turns relevant
real-world evidence into directional hypotheses for a defined population, geography, behavioural
question, and use case. It is not a replacement for real-world activities, but a conduit and
complementary asset to problem-solve faster, more robustly, and with more resource-efficient
approaches. It works very well with established real-world research, data collection, prototyping,
pilots, and at-scale implementation practices.</p>
<p>HIE does not write virtual personas as character descriptions or prompts. It converts
real-world evidence into machine-readable behavioural agents, places them in structured social
and research environments, and runs repeated simulations under explicit rules, constraints and
confidence checks. The result is simulation that is bounded by evidence, persistent in memory
over time and auditable after the fact.</p>

<h2><span class="n">01</span>The evidence base</h2>
<p>Every virtual persona, behaviour, interaction and sandbox in HIE is grounded in real-world
evidence: qualitative research, surveys, population datasets, longitudinal records and other
public and private sources. HIE has an ever-growing evidence corpus.</p>

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</div>

<div class="wide"><div class="stats">

<div class="stat">3.2 BnReal-world research data points<small>Used to ground virtual personas, behaviours, interactions and sandboxes</small></div>

<div class="stat">1.6 MnReal-world participants<small>Represented directly in the underlying observational and research studies</small></div>

<div class="stat">1.67 BnPopulation represented<small>Through weighted cohorts built on that evidence</small></div>

<div class="stat">59.5 BnSimulation data points generated<small>Evaluations, state transitions and simulations run on the platform</small></div>

</div></div>

<div class="wrap">

<p>That evidence is mapped into a common schema. It is never pasted directly into a language model prompt. From there, one logic chain runs through the whole engine:</p>

<ol class="chain" aria-label="HIE logic chain">EvidenceVirtual persona schemaInstantiationRulesInteractionCohortInstrumentSimulationConfidenceValidation

<h2>02How the engine was trained</h2>

<p>HIE began as a shell: machine learning code that defines how a virtual persona is constructed, with metrics across four categories: demographics, psychographics, behaviours and context. On its own, the shell knows the shape of a virtual person but nothing about specific characteristics that define it.</p>

<p>We trained it on a wide range of real-world data, including human-centered design primary research, in-field observations, household and national surveys, census data, quantitative research studies, clinical and programme data, longitudinal series data, and more. Large language models (LLMs) made the training fast and feasible, reading and structuring very different sources and mapping them into the shell's common schema. That structured evidence trained the machine learning code and let us find the right weights, turning a static shell into a dynamic machine learning schema that can generate new virtual personas whenever new data is loaded.</p>

</div>

<div class="wide"><div class="grid3">

<div class="card tint">Data-sets by Dalberg<h4>Engagements over 25 years</h4><p style="margin:0">Dalberg brings over two decades of primary research experience on macroeconomic, demographic, behavioral, and psychographic factors across multiple LMICs. We use the insights to co-design and problem-solve with communities, governments, social/private enterprises, CBOs, and NGOs.</p></div>

<div class="card">National household and health surveys<h4>Population baselines</h4><p style="margin:0">For example, GAVI / UNICEF / WHO datasets, NFHS-5 (India), DHS (Kenya, Nigeria, Ghana, Tanzania, Bangladesh), NAHS (Nigeria), IHDS-II and NSSO 75th Round, plus census and open government data, among others.</p></div>

Health research and clinical evidence

Behaviour in care

For example, IBBSS key population surveillance and PrEP cohort studies (Nigeria), Nivi.io data from Uganda/India etc., the National Mental Health Survey of India, the HAP and Atmiyata community mental health trials, and conversational maternal health data, among others.

Livelihoods and enterprise

Economic lives

For example, DAY-NRLM self-help group registries, the PRADAN women's collectives panel, IWWAGE enterprise diaries, Dalberg's "Becoming Someone" aspiration study, and rural digital enterprise field audits, among others.

Agriculture and climate

Farms and weather

For example, smallholder agronomic and input adoption studies in Kenya, agricultural extension records, and India Meteorological Department climate and heat data, Digital Green data from their primary research, longitudinal ten-year climate-heat series data across countries, among others.

Education

Learning

For example, ASER learning outcomes, alongside national school health and well-being frameworks; Acumen Academy published results, Teach For All and country-level published results, national education surveys, among others.

The above are a few examples of the types of input data we have used to train the model. The Human Insights Engine doesn't retain the prior training data. The training taught us **the quality and quantity of data needed** to generate new virtual personas and virtual worlds for a specific problem. That formula is what let us build demo centres, mini virtual worlds dedicated to specific problem-solving spaces.

This has helped us codify the approach needed to build any new set of virtual personas and virtual worlds for specific problem areas, and do it quickly and efficiently. It tells us what kind of data to collect first. Once that foundation is built, it can be reused to solve many more problems, not just the first one.

Two principles that keep the engine honest

Principle 1

Trained only on inputs

When we use a study, we train only on its original responses, never on the study's outputs or findings.

Principle 2

Simulated data never goes back in

Nothing the engine generates is ever fed back into the platform as training data.

Both guard against the same failure. Feeding in simulated data risks hallucination compounding over time. Feeding in a study's outputs risks the model, over time, referencing those answers instead of reasoning from what people actually said and did.

<p>It also means there is only one route to improving HIE: more detailed, more comprehensive, higher-quality input data. When a simulation needs to be more accurate, we improve what goes in, never by feeding back what comes out.</p>
<div class="callout">Inputs are the past, a study's outputs are the present, and simulations are the future.Trained only on inputs, the engine has no sense of past, present or future. Reproducing a study that has already happened and anticipating one that has not are the same task, which is why HIE's validation results are a strong anchor for trusting its simulations.</div>
<p>The animation below walks the build end to end: from an empty shell, through training and calibration, to hundreds of distinct virtual personas interacting inside dynamic virtual worlds. Use the controls to pause or step through it.</p>
</div>

<figure class="hie" data-hie="1" tabindex="0" aria-label="Animation: how HIE is built, seven steps. Focus and use arrow keys to step, space to pause."></figure>

<div class="wrap">
<h2>03How a virtual persona is built</h2>
<p>A virtual persona in HIE is a structured state model, not a character sketch. Each one is represented through four layers, which together decide what it can be, how it tends to behave, what can change, and where every part of it came from.</p>
</div>
<div class="wide"><div class="grid4">
<div class="card tint">A · Fixed core<h4>Who the virtual persona is</h4><p>The segment boundary it cannot casually drift outside.</p><p class="ex">Geography, age band, household, livelihood, literacy, language, socioeconomic context, structural constraints</p></div>
<div class="card tint">B · Behavioural priors<h4>How it tends to behave</h4><p>Relatively stable tendencies, calibrated as ranges rather than single fixed numbers.</p><p class="ex">Risk tolerance, present bias, agency, norms, stigma, preferences, trust tendencies, channel use, price sensitivity</p></div>
<div class="card tint2">C · Dynamic state<h4>What can change</h4><p>States that evolve as it meets information, people or events, while the fixed core stays intact.</p><p class="ex">Trust, relevance, emotion, intent, perceived risk, feasibility, social permission, salience</p></div>
<div class="card tint2">D · Evidence and memory<h4>Where it came from</h4><p>Links behaviour back to provenance and prior experience, so it is traceable rather than purely generative.</p><p class="ex">Source IDs, sampling frame, recency, evidence limitations and anchors, prior interactions, observed outcomes, trust matrix</p></div>
</div></div>
<div class="wrap">
<h3>From evidence to a calibrated agent</h3>
<p>Building a virtual persona follows five steps: retrieve evidence, map it to the virtual persona schema, calibrate attributes and

behavioural ranges

,

instantiate stochastic latent variation

, and

validate constraints and provenance

.

Calibration is where the shell becomes dynamic, and it never involves feeding in outputs. Simulated behaviour is compared with what was observed in the real world. Where accuracy falls short, the weights for new virtual personas are adjusted by feeding in more detailed, more comprehensive, higher-quality input data until simulations reach the accuracy needed. Once calibrated, the same machinery can take in new data and generate new virtual personas without being rebuilt.

Stochastic Latent Seeding (SLS) is what turns one segment into many distinct virtual personas. The segment defines the behavioural envelope, and SLS decides where a particular virtual persona starts inside it. Two virtual personas from the same segment might begin with slightly different levels of trust, household pressure or risk tolerance, while both staying inside the calibrated evidence boundaries.

Virtual persona schema = what this person can plausibly be. SLS = which plausible version is instantiated for this run.

This is why HIE produces a distribution of plausible responses rather than one deterministic answer.

What a virtual persona is, and is not, allowed to do

- Hard constraints do not drift.** Core demographic and structural attributes stay fixed unless a simulation explicitly models their change.
- Variation stays inside calibrated boundaries.** Stochasticity creates heterogeneity, not arbitrary behaviour.
- Dynamic states change through events and interaction.** Trust, intent, relevance, emotion and feasibility move as the scenario unfolds.
- Evidence gaps remain visible.** Where evidence is thin, the system surfaces uncertainty rather than silently generalising.
- State transitions are traceable.** Changes in state, assumptions and evidence used in a run are logged.
- The model does not redefine the virtual persona.** The underlying language model is frozen and works inside boundaries set by the engine.

04From virtual personas to virtual worlds

Individual virtual personas become useful when they are placed in context. HIE layers four structures on top of them.

The interaction graph governs how virtual personas affect each other. HIE does not simply concatenate responses. Each exchange is interpreted through relationships, trust, authority, social norms, influence and previous interactions, and can shift dynamic states such as trust or social permission.

Units and cohorts assemble virtual personas into households, panels, networks and communities against a sampling frame, with weights so the cohort represents a larger real population. Aggregation does not erase individual state.

The instrument runner applies qualitative, quantitative and specialised research methods across the cohort. A virtual persona can move along a behavioural pathway,

from exposure and attention through comprehension, credibility, personal relevance, social consultation and intent to action or drop-off. HIE records where and why drop-off occurs, not just the final answer.

The sandbox engine is a persistent environment for repeated and counterfactual simulation. Conditions such as price, service availability, policy, weather, rumours or delivery channels can change virtual personas' dynamic states without rewriting the evidence base. The same world can then run scenario A against scenario B, before and after, Monte Carlo runs, sensitivity tests, or a historical replay.

How virtual personas learn during a run

```
Current state receives a stimulus, question or interaction
interprets it through evidence, priors, trust and context
decides or responds
sees an outcome, social response or new information
updates its dynamic state
next decision
```

Virtual personas are context-aware and persistent in memory within a simulation, but the underlying foundation model is never retrained.

The engine decides; the language model speaks

HIE engine determines

Evidence retrieval, virtual persona state, constraints, behavioural parameters, decision pathway, state transitions, sampling and weighting, and simulation conditions.

Frozen language model provides

Situated reasoning, interpretation, natural-language expression and dialogue, using the evidence placed in its context.

A structured decision policy produces the choice or propensity. The language model then explains it in the virtual persona's voice. It expresses the virtual persona; it does not define it.

05 Confidence at four levels

HIE scores confidence at four levels, because a well-grounded virtual persona can still give a weak answer on an unfamiliar topic, and a strong answer can still come from a poorly designed study.

Level 1 · Virtual persona build

Is this virtual persona sufficiently grounded?

Evidence coverage, source quality, schema completeness, recency, calibration quality, provenance and how much inference was needed.

Level 2 · Response

How trustworthy is this specific answer?

Retrieval relevance, evidence availability, state consistency, stability across reruns and unsupported inference.

<div class="card">Level 3 · Panel or study<h4>How reliable is the aggregated finding?</h4><p>Population coverage, sampling-frame fit, weighting, segment representation, method fit, variance across runs and sensitivity to assumptions.</p></div>
<div class="card tint">Level 4 · Virtual World Confidence<h4>Is this simulation fit for this question?</h4><p>Confidence in the quality of the simulation set-up, not the probability that a simulated outcome will occur.</p></div>
</div></div>

<div class="wrap">

<p>The Level 4 score, the Virtual World Confidence, is a weighted composite of six components:</p>

<div class="weights" role="img" aria-label="Weights: evidence quality 30%, virtual persona fidelity 20%, method fit 15%, delivery and execution fidelity 15%, assumptions 10%, audit and validation 10%">

<div style="flex:30;background:#B0164F">30%</div><div style="flex:20;background:#D23F7B">20%</div><div style="flex:15;background:#EC6A9E">15%</div><div style="flex:15;background:#1FA9D6">15%</div><div style="flex:10;background:#0A739A">10%</div><div style="flex:10;background:#4A0F2E">10%</div></div>

<ul class="wlegend">

<i style="background:#B0164F"></i>Evidence quality: strength, relevance, representativeness, recency

<i style="background:#D23F7B"></i>Virtual persona fidelity: how well they represent the population and its heterogeneity

<i style="background:#EC6A9E"></i>Method fit: whether the method suits the question

<i style="background:#1FA9D6"></i>Delivery and execution fidelity: channels, stimuli and journeys realistically represented

<i style="background:#0A739A"></i>Assumptions: importance and uncertainty of what is not grounded in evidence

<i style="background:#4A0F2E"></i>Audit and validation: traceability, stability tests, backward validation, expert review

<h2>06Validated against real-world studies</h2>

<p>The clearest test of HIE is backward validation: give the engine only the inputs of a completed real-world study, and compare what its virtual personas say or do with what real people said or did. Across four exercises spanning agriculture, financial inclusion and health, HIE reached 94% to 99.5% benchmark similarity, with a substantially lower sample size and over multiple runs of the same study.</p>

</div>

<figure class="hie" data-hie="2" tabindex="0" aria-label="Animation: validation results and Virtual World Confidence. Focus and use arrow keys to step, space to pause."></figure>

<div class="wide"><div class="studies">

<div class="card">Kenya · Farmers95%<p>Digital Green, FarmerChat survey by 60 Decibels. Roughly 2–8% variance from real-world findings across tested survey measures.</p></div>

<div class="card">India · Mothers and frontline workers97%<p>Government of Maharashtra, NFHS-5 antenatal care. 31,643 households, 33,755 women and 5,497 men on comparable ANC indicators.</p></div>

<div class="card">Bangladesh · National population94%<p>BRAC, nationwide study on financial behaviours. 3,592 respondents across all eight divisions and six behavioural segments.</p></div>

<div class="card">Nigeria · Female sex workers99%<p>Lagos University Teaching Hospital, HIV PrEP study. 310 respondents; mean positive response 59.35% human versus 59.83% simulated.</p></div></div>

<div class="wrap">

<p>Backward validation sits alongside five other checks: holdout comparison against real observations not used to build the virtual personas; repeated-run stability across many SLS instantiations; sensitivity testing of assumptions, weights and conditions; expert review of behavioural mechanisms and delivery assumptions; and field calibration, where small real-world samples of original responses are added as new input to recalibrate parameters without replacing the evidence base.</p>

</div>

<div class="wide" style="max-width:808px"><div class="cols">

<div class="card tint">Remains controlled and versioned<p style="margin:0">Source evidence, virtual persona schema, fixed demographic core, structural constraints, model weights, simulation rules, provenance, sampling frame.</p></div>

<div class="card tint2">Can change during a simulation<p style="margin:0">Trust, relevance, emotion, intent, perceived risk, feasibility, social permission, memory and interaction history.</p></div>

</div></div>

<div class="wrap">

<h2>07What this means for decisions</h2>

<p>HIE lets a team test a policy, program, product, service, or message against thousands of evidence-grounded virtual personas before committing resources, see where and why people would drop off, and compare scenarios side by side. Its outputs are directional hypotheses with an explicit confidence score and a traceable evidence trail. They sharpen where to invest, what to test in the field, and what to ask real people next, rather than replacing them.</p>

<div class="foot">Dalberg · Human Insights Engine · Confidential. Animation labels marked "Illustrative" show example variables, outcomes and sectors, not specific model outputs.</div>

</div>

</main>

<script>

(()=>{

```

let X=null,SC=1;
const
INK='#4A0F2E',MUT='#6E5664',LN='#E6D5DE',K=['#C2185B','#EC6A9E','#E39B2D','#1FA9D6'
],KT=['#B0164F','#C23A78','#A8660F','#0A739A'],WH='#FFFFFF';
const DF=""'Funnel Display','Plus Jakarta Sans',system-ui,sans-serif",BF=""'Plus Jakarta
Sans',system-ui,sans-serif";
const TAU=Math.PI*2,cl=v=>v<0?0:v>1?1:v,E=v=>{v=cl(v);return
v*v*(3-2*v)},EO=v=>{v=cl(v);return 1-Math.pow(1-v,3)},L=(a,b,t)=>a+(b-a)*t;
let sd=12345;const R=(s)=>(sd=(sd*16807)%2147483647)/2147483647;
const hx=h=>[1,3,5].map(i=>parseInt(h.slice(i,i+2),16));
const rgba=(h,a)=>{const c=hx(h);return `rgba(${c[0]},${c[1]},${c[2]},${a})`;};
const mix=(a,b,t)=>{const A=hx(a),B=hx(b);return
`rgb(${A.map((v,i)=>Math.round(v+(B[i]-v)*t)).join(',')})`;};

```

// drawing helpers

```

function T(s,x,y,z,c,al,w,a,f){X.globalAlpha=a==null?1:a;X.font=`$w||500} $z}px
${f||BF}`;X.fillStyle=c;X.textAlign=al||'left';X.fillText(s,x,y);X.globalAlpha=1}
function
D(x,y,r,c,a){X.globalAlpha=a==null?1:a;X.fillStyle=c;X.beginPath();X.arc(x,y,r,0,TAU);X.fill();X.glo
balAlpha=1}
function
Ln(x1,y1,x2,y2,c,w,a,dash){X.globalAlpha=a==null?1:a;X.strokeStyle=c;X.lineWidth=w;X.lineCa
p='round';X.setLineDash(dash||[]);X.beginPath();X.moveTo(x1,y1);X.lineTo(x2,y2);X.stroke();X.s
etLineDash([]);X.globalAlpha=1}
function
glass(x,y,w,h,r,a){if(a<=0)return;X.save();X.globalAlpha=a;X.beginPath();X.roundRect(x,y,w,h,r);
const
g=X.createLinearGradient(0,y,0,y+h);g.addColorStop(0,'rgba(255,255,255,.96)');g.addColorStop
(1,'rgba(255,255,255,.88)');X.fillStyle=g;X.shadowColor='rgba(74,15,46,.12)';X.shadowBlur=30*
SC;X.shadowOffsetY=10*SC;X.fill();X.shadowColor='transparent';X.strokeStyle='rgba(74,15,46,.
07)';X.lineWidth=1.5;X.stroke();
X.beginPath();X.moveTo(x+r,y+1);X.lineTo(x+w-r,y+1);X.restore()}
function
pill(x,y,w,h,fill,stroke,a){X.save();X.globalAlpha=a;X.beginPath();X.roundRect(x,y,w,h,h/2);X.fillSt
yle=fill;X.fill();if(stroke){X.strokeStyle=stroke;X.lineWidth=2;X.stroke();X.restore()}
function
curve(x1,y1,x2,y2,bend,c,w,a,dash){X.save();X.globalAlpha=a;X.strokeStyle=c;X.lineWidth=w;X.
setLineDash(dash||[]);X.beginPath();X.moveTo(x1,y1);X.bezierCurveTo(L(x1,x2,.5),y1+bend,L(x
1,x2,.5),y2-bend,x2,y2);X.stroke();X.restore()}
function onCurve(x1,y1,x2,y2,bend,f){const
u=1-f,cx1=L(x1,x2,.5),cy1=y1+bend,cx2=cx1,cy2=y2-bend;
return[u*u*u*x1+3*u*u*f*cx1+3*u*f*f*cx2+f*f*f*x2,u*u*u*y1+3*u*u*f*cy1+3*u*f*f*cy2+f*f*f*y2]}

```

```
// ----- the person -----
```

```
const
```

```
HL=[[0,18],[-32,24],[-52,50],[-58,90],[-55,126],[-43,152],[-28,168],[-26,194],[-54,210],[-104,222],[-134,240],[-148,276],[-153,332],[-157,392],[-161,446],[-165,502],[-167,548],[-173,584],[-170,622],[-154,640],[-140,616],[-136,570],[-129,512],[-121,452],[-113,396],[-105,334],[-99,300],[-95,360],[-87,430],[-92,482],[-104,532],[-106,604],[-98,684],[-90,762],[-92,832],[-80,918],[-73,946],[-93,968],[-62,982],[-30,976],[-27,946],[-33,862],[-29,780],[-19,690],[-9,612],[0,578]];
```

```
const FP=HL.concat(HL.slice(1,-1).reverse().map([(x,y)]=>[-x,y]));
```

```
function cr(pts){const p=new Path2D(),n=pts.length;p.moveTo(pts[0][0],pts[0][1]);
```

```
for(let i=0;i<n;i++){const
```

```
a=pts[(i-1+n)%n],b=pts[i],c=pts[(i+1)%n],d=pts[(i+2)%n];p.bezierCurveTo(b[0]+(c[0]-a[0])/6,b[1]+(c[1]-a[1])/6,c[0]+(d[0]-b[0])/6,c[1]+(d[1]-b[1])/6,c[0],c[1]);p.closePath();return p}
```

```
const FIG=cr(FP);let PER=0;for(let i=0;i<FP.length;i++){const
```

```
a=FP[i],b=FP[(i+1)%FP.length];PER+=Math.hypot(a[0]-b[0],a[1]-b[1]);PER*=1.1;
```

```
const cat=(x,y)=>y<172?1:y>585?3:(Math.abs(x)>108&&y>232)?2:0; // 0 demographics,1
```

```
psychographics,2 behaviours,3 context
```

```
const tc=document.createElement('canvas').getContext('2d');
```

```
const SL=[];for(let row=0;for(let y=22;y<990;y+=12.5,row++)for(let
```

```
x=-178+(row%2)*6.5;x<178;x+=13){const
```

```
jx=x+(R()-0.5)*3,jy=y+(R()-0.5)*3;if(tc.isPointInPath(FIG,jx,jy))SL.push({x:jx,y:jy,c:cat(jx,jy)}}}
```

```
const BIG={cx:960,cy:640,h:580};
```

```
SL.forEach(s=>{const k=BIG.h/1000;s.X=BIG.cx+s.x*k;s.Y=BIG.cy+(s.y-500)*k;const
```

```
side=R()*0.5;s.sx=(side?300:1620)+(R()-0.5)*170;s.sy=560+R()*300;s.d=(1-s.y/1000)*.32+R()*0.34;s.tw=R()*TAU});
```

```
const toS=(ux,uy,f)=>{f=f||BIG;const k=f.h/1000;return[f.cx+ux*k,f.cy+(uy-500)*k]};
```

```
function figLive(cx,cy,h,a,flick,t){const
```

```
k=h/1000;X.save();X.translate(cx,cy);X.scale(k,k);X.translate(0,-500);X.globalAlpha=a;
```

```
for(let ci=0;ci<4;ci++){X.fillStyle=K[ci];X.beginPath();for(const s of
```

```
SL)if(s.c===ci){X.moveTo(s.x+4.6,s.y);X.arc(s.x,s.y,4.6,0,TAU)}X.fill()}
```

```
if(flick>0){X.fillStyle=WH;X.beginPath();for(let i=0;i<SL.length;i+=3){const
```

```
s=SL[i];if(Math.sin(t*.011+s.tw*9)>.85){X.moveTo(s.x+5,s.y);X.arc(s.x,s.y,5,0,TAU)}}X.globalAlpha=a*flick;X.fill()}
```

```
X.globalAlpha=a*.7;X.strokeStyle=INK;X.lineWidth=2/k;X.stroke(FIG);X.restore();X.globalAlpha=1}
```

```
function sprite(tint,amt){const
```

```
q=.72,c=document.createElement('canvas');c.width=Math.round(380*q);c.height=Math.round(1000*q);const x=c.getContext('2d');x.scale(q,q);x.translate(190,0);
```

```
const cols=K.map(k=>tint===null?k:mix(k,K[tint],amt));
```

```
for(let ci=0;ci<4;ci++){x.fillStyle=cols[ci];x.beginPath();for(const s of
```

```
SL)if(s.c===ci){x.moveTo(s.x+5.4,s.y);x.arc(s.x,s.y,5.4,0,TAU)}x.fill()}
```

```
x.strokeStyle='rgba(74,15,46,.5)';x.lineWidth=4;x.stroke(FIG);return c}
```

```

const
SP=[sprite(null,0),sprite(0,.5),sprite(1,.5),sprite(2,.5),sprite(3,.5),sprite(0,.8),sprite(3,.75),sprite(1,
.75)];
function spr(i,cx,by,h,wm,a){const
w=h*.38*(wm||1);X.globalAlpha=a==null?1:a;X.drawImage(SP[i],cx-w/2,by-h*.985,w,h);X.global
Alpha=1}
function ground(cx,y,rx,a){X.save();X.globalAlpha=a;const
g=X.createRadialGradient(cx,y,0,cx,y,rx);g.addColorStop(0,'rgba(194,24,91,.16)');g.addColorSto
p(1,'rgba(194,24,91,0)');X.fillStyle=g;X.beginPath();X.ellipse(cx,y,rx,rx*.16,0,0,TAU);X.fill();X.rest
ore()}
function halo(cx,cy,r,col,a){X.save();X.globalAlpha=a;const
g=X.createRadialGradient(cx,cy,0,cx,cy,r);g.addColorStop(0,rgba(col,.22));g.addColorStop(1,rgb
a(col,0));X.fillStyle=g;X.beginPath();X.arc(cx,cy,r,0,TAU);X.fill();X.restore()}

// ----- slide 1 stages -----
const CO=[{c:1,n:'Psychographics',v:'Values, aspirations, trust, risk
appetite',side:-1,ly:420,a:[-30,95]},
{c:0,n:'Demographics',v:'Age, gender, income, household, location',side:-1,ly:650,a:[-55,400]},
{c:2,n:'Behaviours',v:'Habits, spending, channel and service use',side:1,ly:560,a:[150,592]},
{c:3,n:'Context',v:'Access, social norms, infrastructure',side:1,ly:800,a:[52,935]}];
function outline(p,t,glow){const
k=BIG.h/1000;X.save();X.translate(BIG.cx,BIG.cy);X.scale(k,k);X.translate(0,-500);
X.setLineDash([PER*p,PER]);X.strokeStyle=INK;X.lineWidth=2.6/k;X.shadowColor='rgba(31,16
9,214,.55)';X.shadowBlur=(glow||14)*SC;X.stroke(FIG);X.setLineDash([]);X.shadowColor='trans
parent';X.restore()}
function S_shell(p,t){const
k=BIG.h/1000;ground(BIG.cx,BIG.cy+BIG.h/2+4,190,E(p*3));halo(BIG.cx,BIG.cy-40,380,K[3],.5*
E(p*2));
const sa=E((p-.3)/.3)*.2;if(sa>0){X.fillStyle=rgba(INK,sa);X.beginPath();for(const s of
SL){X.moveTo(s.X+1.7,s.Y);X.arc(s.X,s.Y,1.7,0,TAU)}X.fill()}
outline(EO(p/.5),t);
const
sc=E((p-.45)/.2);if(sc>0){X.save();X.translate(BIG.cx,BIG.cy);X.scale(k,k);X.translate(0,-500);X.c
lip(FIG);const
yy=((t*.00028)%1)*1100-50,g=X.createLinearGradient(0,yy-60,0,yy+60);g.addColorStop(0,'rgba(
31,169,214,0)');g.addColorStop(.5,'rgba(31,169,214,${.3*sc})');g.addColorStop(1,'rgba(31,169,
214,0)');X.fillStyle=g;X.fillRect(-200,yy-60,400,120);X.restore()}
CO.forEach((o,i)=>{const a=E((p-.38-i*.1)/.22);if(a<=0)return;const
[ax,ay]=toS(o.a[0],o.a[1]),lx=o.side<0?716:1204,ly=o.ly-10,ex=ax+o.side*46;
X.save();X.globalAlpha=a;X.strokeStyle=rgba(K[o.c],.8);X.lineWidth=1.8;X.beginPath();X.moveT
o(lx,ly);X.lineTo(L(lx,ex,a),ly);if(a>.6){const
b=(a-.6)/.4;X.lineTo(L(ex,ax,b),L(ly,ay,b))}X.stroke();X.restore();
D(ax,ay,6+2*Math.sin(t*.005+i),K[o.c],a);halo(ax,ay,34,K[o.c],a);

```

```

T(o.n,o.side<0?700:1220,o.ly,36,KT[o.c],o.side<0?'right':'left',700,a,DF);T(o.v,o.side<0?700:1220,o.ly+36,22,MUT,o.side<0?'right':'left',500,a)))}
function counters(p,a){const m=EO(p/.75);
T((1.6*m).toFixed(1)+'M',300,480,104,INK,'center',700,a,DF);T('real-world
participants',300,522,24,MUT,'center',600,a);T('observational and research
evidence',300,554,20,MUT,'center',500,a*.8);
T((3.2*m).toFixed(1)+'B',1620,480,104,INK,'center',700,a,DF);T('real-world
research',1620,522,24,MUT,'center',600,a);T('data points',1620,554,20,MUT,'center',500,a*.8)}
function S_train(p,t){const k=BIG.h/1000;ground(BIG.cx,BIG.cy+BIG.h/2+4,190,1);let filled=0;
const fa=E(p*.4);counters(p,fa);
X.fillStyle=rgba(INK,.16);X.beginPath();for(const s of
SL){if(p<s.d){X.moveTo(s.X+1.7,s.Y);X.arc(s.X,s.Y,1.7,0,TAU)}}X.fill();
for(let ci=0;ci<4;ci++){X.fillStyle=K[ci];X.beginPath();for(const s of
SL){if(s.c===ci&&p>=s.d+.2){filled++;X.moveTo(s.X+2.9,s.Y);X.arc(s.X,s.Y,2.9,0,TAU)}}X.fill()}
halo(BIG.cx,BIG.cy-40,420,K[0],.5*filled/SL.length);
for(const s of SL){const u=(p-s.d)/.2;if(u<=0||u>=1)continue;for(let q=0;q<3;q++){const
uu=EO(u*q*.05);if(uu<=0)continue;const
f=uu,x=(1-f)*(1-f)*s.sx+2*(1-f)*f*L(s.sx,s.X,.5)+f*f*s.X,y=(1-f)*(1-f)*s.sy+2*(1-f)*f*(Math.min(s.sy,s
.Y)-160)+f*f*s.Y;D(x,y,q?1.6:2.6,K[s.c],q?.35/q:1)}}
outline(1,t,10+10*filled/SL.length)}
function S_seg(p,t){const
m=EO(p/.38),fx=L(BIG.cx,300,m),fh=L(BIG.h,440,m),fy=L(BIG.cy,640,m);ground(fx,fy+fh/2+4,L(
190,130,m),1);figLive(fx,fy,fh,1,0,t);
T('Trained shell',fx,fy+fh/2+52,26,MUT,'center',600,E((p-.3)/.2));
for(let c=0;c<3;c++){const a=EO((p-.26-c*.12)/.24);if(a<=0)continue;const
x=610+c*410,y=L(360,330,a),w=370,h=560;
curve(fx+90,fy-40,x,y+h/2,0,rgba(K[3],.6),2,a*.7,[4,8]);const
f=((t*.0005)+c*.3)%1,[qx,qy]=onCurve(fx+90,fy-40,x,y+h/2,0,f);D(qx,qy,4,K[3],a*Math.sin(f*Math.
PI));
glass(x,y,w,h,22,a);T('Region '+ (c+1),x+30,y+58,32,INK,'left',700,a,DF);T('Virtual persona
segments',x+30,y+92,21,MUT,'left',600,a);
for(let j=0;j<3;j++){const px=x+72+j*113,si=1+((c*3+j)%7),hh=270;spr(si,px,y+h-120,hh,1,a);
let bx=px-46;const
wts=[.2+((c+j)*37%10)/20,.15+((c*7+j*3)%10)/25,.1+((c*5+j*9)%10)/22,.12+((c+j*4)%10)/28],tot
=wts.reduce((u,v)=>u+v,0);
wts.forEach((wv,wi)=>{const
ww=92*wv/tot;X.globalAlpha=a;X.fillStyle=K[wi];X.fillRect(bx,y+h-84,ww-2,8);bx+=ww});X.global
Alpha=1;
T('Segment '+String.fromCharCode(65+c*3+j),px,y+h-40,20,MUT,'center',600,a)}}}
const WN=['Income','Trust','Risk appetite','Spending','Access','Social
norms'],ON=['Adoption','Spend','Uptake','Retention','Trust','Referral'];
const WT=[.62,.35,.74,.48,.56,.3],RV=[.78,.52,.66,.9,.44,.6],ER=[-.3,.3,-.22,-.38,.36,.25];
function S_cal(p,t){const
s=E(p/.7),d=E((p-.66)/.18),cx=960,cy=590,h=480;ground(cx,cy+h/2+4,170,1);

```

```

glass(110,330,500,570,22,1);T('Model weights',142,386,30,INK,'left',700,1,DF);T('Illustrative
variables',142,420,21,KT[0],'left',600);
glass(1310,330,500,570,22,1);T('Simulated vs
real-world',1342,386,30,INK,'left',700,1,DF);T('Illustrative
outcomes',1342,420,21,KT[0],'left',600);
for(let i=0;i<6;i++){const
y=488+i*62,v=WT[i]+.34*(1-s)*Math.sin(t*.005+i*1.9),kx=L(142,578,cl(v));T(WN[i],142,y-14,21,M
UT,'left',600);
Ln(142,y+6,578,y+6,LN,6);Ln(142,y+6,kx,y+6,K[i%4],6);D(kx,y+6,11,INK);
T(ON[i],1342,y-14,21,MUT,'left',600);const
rv=RV[i],sv=rv+ER[i]*(1-s);X.strokeStyle='rgba(74,15,46,.45)';X.lineWidth=2;X.strokeRect(1342,
y-2,430*rv,18);X.fillStyle=K[3];X.fillRect(1342,y+1,430*cl(sv),12)}
X.strokeStyle='rgba(74,15,46,.45)';X.lineWidth=2;X.strokeRect(1342,862,24,12);T('Real-world',1
376,875,20,MUT,'left',600);X.fillStyle=K[3];X.fillRect(1510,862,24,12);T('Simulated',1544,875,20,
MUT,'left',600);
[[610,590,720],[1200,590,1310]].forEach(([a,b,c2],j)=>{Ln(a,b,c2,b,LN,2,1,[3,7]);for(let
q=0;q<3;q++){const f=(t*.0006+q/3)%1;D(L(a,c2,f),b,4.5,j?K[3]:K[2],Math.sin(f*Math.PI))});
halo(cx,cy-30,360,d>0?K[3]:K[1],.35+.35*d);figLive(cx,cy,h,1,1-s,t);
X.save();X.translate(cx,cy-20);X.rotate(t*.0003);X.strokeStyle=d>0?rgba(K[3],.35+.5*d):'rgba(11
0,86,100,.35)';X.lineWidth=2.5;X.setLineDash([2,14]);X.beginPath();X.ellipse(0,0,225,275,0,0,TA
U);X.stroke();X.rotate(-t*.0006);X.setLineDash([30,18]);X.globalAlpha=d;X.beginPath();X.ellipse(
0,0,240,292,0,0,TAU);X.stroke();X.restore();
if(d>0){const w=620;pill(cx-w/2,872,w,52,WH,K[3],d);T('Dynamic model: new data in, new virtual
personas out',cx,906,21,INK,'center',700,d)}}
const CR=[];for(let r=0;r<7;r++){const n=26-r*2,yb=520+r*62,h=80+r*20;for(let
i=0;i<n;i++){CR.push({x:640+(i+.5)*(1160/n)+(R()-5)*18,yb,h:h*(.9+R()*.2),wm:.85+R()*.35,sp:M
ath.floor(R()*8),born:.12+r*.05+R()*.33,a:.5+r*.08,ph:R()*TAU})}
function S_people(p,t){glass(110,330,420,580,22,1);T('One
segment',142,390,30,INK,'left',700,1,DF);spr(1,320,690,260,1,1);
T('Vary each variable slightly',142,738,21,MUT,'left',600);
['Demographics','Psychographics','Behaviours','Context'].forEach((n,i)=>{const
y=778+i*34,kx=425+34*Math.sin(t*.018+i*2.1)*E(p*3);Ln(350,y,500,y,LN,5);D(kx,y,8,K[i]);T(n,14
2,y+7,20,MUT,'left',600));
for(let q=0;q<4;q++){const f=(t*.0007+q/4)%1;D(L(540,640,f),620,4,K[q],Math.sin(f*Math.PI))}
ground(1220,896,640,E(p*3));
for(const c of CR){const
a=E((p-c.born)/.1);if(a<=0)continue;spr(c.sp,c.x+Math.sin(t*.0012+c.ph)*2,c.yb+(1-a)*24,c.h,c.w
m,a*c.a)}
T('Hundreds of virtual personas, each slightly
different',1220,420,30,INK,'center',700,E((p-.45)/.25),DF)}
const G=[[700,590],[1200,590],[760,830],[1240,830]];
const PX=[];for(let i=0;i<44;i++){const
a=R()*TAU,r=Math.sqrt(R());PX.push({sx:540+R()*880,sy:470+R()*400,g:i%4,ox:Math.cos(a)*r*1
20,oy:Math.sin(a)*r*60,sp:1+(i%4===0?0:i%4===1?1:i%4===2?2:3),ph:R()*TAU})}

```

```

const
IN=[['Money',205,400],['Bank',205,540],['Home',205,680],['App',205,820],['Clinic',1715,440],['Hospital',1715,600],['Vaccine',1715,760]];
function
icon(n,x,y,c){X.save();X.strokeStyle=c;X.fillStyle=c;X.lineWidth=3;X.lineJoin='round';X.lineCap='round';X.beginPath();
if(n==='Money'){X.roundRect(x-28,y-17,56,34,5);X.moveTo(x+9,y);X.arc(x,y,9,0,TAU)}
else
if(n==='Bank'){X.moveTo(x-30,y-8);X.lineTo(x,y-26);X.lineTo(x+30,y-8);X.closePath();[-18,0,18].forEach(o=>{X.moveTo(x+o,y-2);X.lineTo(x+o,y+16)});X.moveTo(x-32,y+22);X.lineTo(x+32,y+22)}
else
if(n==='Home'){X.moveTo(x-28,y-2);X.lineTo(x,y-26);X.lineTo(x+28,y-2);X.rect(x-20,y-4,40,28);X.rect(x-6,y+8,12,16)}
else if(n==='App'){X.roundRect(x-16,y-27,32,54,7);X.moveTo(x+3,y+18);X.arc(x,y+18,3,0,TAU)}
else
if(n==='Clinic'){X.roundRect(x-24,y-24,48,48,10);X.moveTo(x,y-12);X.lineTo(x,y+12);X.moveTo(x-12,y);X.lineTo(x+12,y)}
else
if(n==='Hospital'){X.rect(x-24,y-24,48,50);X.moveTo(x-9,y-12);X.lineTo(x-9,y+12);X.moveTo(x+9,y-12);X.lineTo(x+9,y+12);X.moveTo(x-9,y);X.lineTo(x+9,y)}
else
if(n==='Vaccine'){X.translate(x,y);X.rotate(-Math.PI/4);X.rect(-8,-20,16,32);X.moveTo(0,-20);X.lineTo(0,-30);X.moveTo(-9,-30);X.lineTo(9,-30);X.moveTo(0,12);X.lineTo(0,28);X.moveTo(-8,-6);X.lineTo(0,-6);X.moveTo(-8,4);X.lineTo(0,4)}
X.stroke();X.restore()}
function S_int(p,t){const m=E((p-.22)/.3),b=E((p-.58)/.2);
const
pos=PX.map((q,i)=>[L(q.sx,G[q.g][0]+q.ox,m)+Math.sin(t*.0015+q.ph)*6,L(q.sy,G[q.g][1]+q.oy,m)+Math.cos(t*.0012+q.ph)*4]);
G.forEach((g,k)=>{if(m<=0)return;halo(g[0],g[1]-30,190,K[k],m);X.save();X.globalAlpha=m*.7;X.strokeStyle=K[k];X.lineWidth=2;X.setLineDash([6,9]);X.lineDashOffset=-t*.02;X.beginPath();X.ellipse(g[0],g[1]-28,170,100,0,0,TAU);X.stroke();X.restore()});
const la=E(p/.15)*.4;for(let i=0;i<22;i++){const
A=pos[i],B=pos[i+22];Ln(A[0],A[1]-36,B[0],B[1]-36,INK,1.4,la*.7*(1-.6*m))}
IN.forEach((n,j)=>{const
on=b>0,a=.55+.45*b;glass(n[1]-85,n[2]-58,170,116,18,a);icon(n[0],n[1],n[2]-12,on?K[0]:MUT);T(n[0],n[1],n[2]+42,21,INK,'center',600,a));
if(b>0)for(let i=0;i<44;i+=2){const
n=IN[i%7],ex=n[1]>900?n[1]-85:n[1]+85,A=pos[i];X.save();X.globalAlpha=b*.35;X.strokeStyle=K[2];X.lineWidth=1.3;X.beginPath();X.moveTo(A[0],A[1]-36);X.quadraticCurveTo(L(A[0],ex,.5),Math.min(A[1],n[2])-60,ex,n[2]);X.stroke();X.restore();

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const
f=(t*.0006+i*.13)%1,u=1-f,qx=u*u*A[0]+2*u*f*L(A[0],ex,.5)+f*f*ex,qy=u*u*(A[1]-36)+2*u*f*(Math.
min(A[1],n[2])-60)+f*f*n[2];D(qx,qy,3.6,K[2],b)}
pos.map((q,i)=>[q,i]).sort((a,b2)=>a[0][1]-b2[0][1]).forEach(([q,i])=>spr(PX[i].sp,q[0],q[1],78,1,1)))
// globe
const FB=[];{const n=760,ga=Math.PI*(3-Math.sqrt(5));for(let i=0;i<n;i++){const
y=1-(i/(n-1))*2,r=Math.sqrt(1-y*y),th=ga*i;FB.push([Math.cos(th)*r,y,Math.sin(th)*r])}}
const vec=(la,lo)=>[Math.cos(la)*Math.sin(lo),Math.sin(la),Math.cos(la)*Math.cos(lo)];
function mkWorld(nc,per,seedShift){const cls=[];for(let i=0;i<nc;i++){const
la=(R()*1.7-.8),lo=R()*TAU,c=Math.floor(R()*4),mem=[];for(let
j=0;j<per;j++)mem.push(vec(la+(R()-.5)*.2,lo+(R()-.5)*.26));cls.push({v:vec(la,lo),c,mem})}
const arcs=[];for(let i=0;i<nc;i++){const
j=(i+1+Math.floor(R()*(nc-1)))%nc;arcs.push([cls[i].v,cls[j].v,R()])}return{cls,arcs}}
const WM=mkWorld(15,24),MINI=[0,1,2,3,4,5].map(()=>mkWorld(6,10));
function proj(v,rot,tilt){const
cr_=Math.cos(rot),sr=Math.sin(rot),x=v[0]*cr_+v[2]*sr,z=-v[0]*sr+v[2]*cr_,ct=Math.cos(tilt),st=Mat
h.sin(tilt);return[x,v[1]*ct-z*st,v[1]*st+z*ct]}
function slerp(a,b,f){const
d=Math.max(-1,Math.min(1,a[0]*b[0]+a[1]*b[1]+a[2]*b[2])),om=Math.acos(d)||1e-6,s=Math.sin(o
m),A=Math.sin((1-f)*om)/s,B=Math.sin(f*om)/s;return[a[0]*A+b[0]*B,a[1]*A+b[1]*B,a[2]*A+b[2]*B]
}
function globe(W,cx,cy,r,rot,tilt,a,t,lat,dom){if(a<=0)return;X.save();X.globalAlpha=a;
halo(cx,cy,r*1.5,K[3],.55);const
g=X.createRadialGradient(cx-r*.38,cy-r*.42,r*.1,cx,cy,r);g.addColorStop(0,'#FFFFFF');g.addColo
rStop(.65,'#EAF4F8');g.addColorStop(1,'#CFE5EE');X.fillStyle=g;X.beginPath();X.arc(cx,cy,r,0,T
AU);X.fill();
X.fillStyle=INK;for(let i=0;i<FB.length;i+=lat){const
q=proj(FB[i],rot,tilt);if(q[2]<=0)continue;X.globalAlpha=a*(.08+.22*q[2]);X.beginPath();X.arc(cx+q
[0]*r,cy-q[1]*r,r>150?1.7:1.1,0,TAU);X.fill()}
W.arcs.forEach((A,i)=>{let
prev=null;X.strokeStyle=rgba(K[2],.55);X.lineWidth=r>150?1.6:1;for(let s=0;s<=24;s++){const
v=slerp(A[0],A[1],s/24),lift=1+.2*Math.sin(Math.PI*s/24),q=proj(v,rot,tilt),P=[cx+q[0]*r*lift,cy-q[1]*r
*lift,q[2]];if(prev&&P[2]>-.05&&prev[2]>-.05){X.globalAlpha=a*.55*cl(P[2]+.4);X.beginPath();X.mo
veTo(prev[0],prev[1]);X.lineTo(P[0],P[1]);X.stroke()}prev=P}
const
f=(t*.00035+A[2])%1,v=slerp(A[0],A[1],f),lift=1+.2*Math.sin(Math.PI*f),q=proj(v,rot,tilt);if(q[2]>0){
X.globalAlpha=a;X.fillStyle=K[2];X.beginPath();X.arc(cx+q[0]*r*lift,cy-q[1]*r*lift,r>150?3.4:2.2,0,T
AU);X.fill()}};
W.cls.forEach(C=>{const
col=dom==null?K[C.c]:mix(K[C.c],K[dom],.55);X.fillStyle=col;C.mem.forEach(m=>{const
q=proj(m,rot,tilt);if(q[2]<=0)return;X.globalAlpha=a*(.25+.75*q[2]);X.beginPath();X.arc(cx+q[0]*r,
cy-q[1]*r,r>150?2.7:1.8,0,TAU);X.fill()}});
X.globalAlpha=a;const
rim=X.createRadialGradient(cx,cy,r*.82,cx,cy,r*1.04);rim.addColorStop(0,'rgba(31,169,214,0)');ri

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m.addColorStop(1,'rgba(31,169,214,.45)');X.fillStyle=rim;X.beginPath();X.arc(cx,cy,r*1.04,0,TAU
);X.fill();X.restore())
const
MP=[[330,410],[240,600],[330,790],[1590,410],[1680,600],[1590,790]],MN=['Health','Agriculture',
'Financial services','Extreme heat','Education','Livelihoods'];
function S_worlds(p,t){const a=EO(p/.22),r=220*L(.6,1,a),cx=960,cy=645;
MP.forEach((m,i)=>{const b=EO((p-.3-i*.07)/.16);if(b<=0)return;const
sx=m[0]<960?cx-r*.9:cx+r*.9;curve(sx,cy,m[0],m[1],0,rgba(K[3],.5),1.6,b*.8,[3,8]);const
f=((t*.0004)+i*.17)%1,[qx,qy]=onCurve(sx,cy,m[0],m[1],0,f);D(qx,qy,3.6,K[3],b*Math.sin(f*Math.PI
))});
globe(WM,cx,cy,r,t*.00016,.36,a,t,1,null);T('Baseline
world',cx,cy+r+56,28,INK,'center',700,a,DF);
MP.forEach((m,i)=>{const
b=EO((p-.3-i*.07)/.16);if(b<=0)return;globe(MINI[i],m[0],m[1],76*L(.3,1,b),t*.00022*(1+i*.15)+i*.3
+i*.05,b,t,4,i%4);T('Illustrative',m[0],m[1]+100,19,KT[0],'center',600,b);T(MN[i],m[0],m[1]+126,22,
INK,'center',700,b));
const
g5=EO((p-.4)/.45);T((59.5*g5).toFixed(1)+'B',cx,354,60,KT[3],'center',700,E((p-.4)/.1),DF);T('sim
ulation data points generated',cx,386,22,MUT,'center',600,E((p-.4)/.1))}

// ----- slide 2 stages -----
const ST4=[{v:95,co:'Kenya',sg:'Farmers',src:'Digital Green, FarmerChat survey by 60
Decibels',d:'Roughly 2–8% variance from real-world findings across the tested survey
measures.'},
{v:97,co:'India',sg:'Mothers and frontline workers',src:'Government of Maharashtra, NFHS-5
antenatal care',d:'31,643 households, 33,755 women and 5,497 men on directly comparable
ANC indicators.'},
{v:94,co:'Bangladesh',sg:'National population',src:'BRAC, nationwide study on financial
behaviours',d:'3,592 respondents across all eight divisions and six established behavioural
segments.'},
{v:99,co:'Nigeria',sg:'Female sex workers',src:'Lagos University Teaching Hospital, HIV PrEP
study',d:'310 respondents. Mean positive response 59.35% human versus 59.83% simulated.'}];
function wrap(s,x,y,mw,lh,z,c,w,a,f){X.font=`$w||500 $zpx $f||BF`;const ws=s.split(' ');let
line="",yy=y;ws.forEach(wd=>{const tt=line?line+
'+wd:wd;if(X.measureText(tt).width>mw&&line)T(line,x,yy,z,c,'left',w,a,f);X.font=`$w||500 $
zpx $f||BF`;line=wd;yy+=lh}else line=tt});if(line)T(line,x,yy,z,c,'left',w,a,f);return yy}
function S_acc(p,t){const chips=[['Study inputs',KT[3]],['HIE virtual personas',K[1]],['Simulated
results',K[0]],['Real-world results',MUT]];let x=110;
X.font=`600 22px $f||BF`;chips.forEach((c,i)=>{const
w=X.measureText(c[0]).width+64,a=EO(p*5-i*.5);pill(x,322,w,56,WH,c[1],a);D(x+26,350,6,c[1],a)
;T(c[0],x+42,358,22,INK,'left',600,a);
if(i<3)T(i===2?'vs':'→',x+w+36,360,24,MUT,'center',600,a);x+=w+72});
ST4.forEach((o,i)=>{const a=EO((p-.14-i*.1)/.25);if(a<=0)return;const
cx0=100+i*435,y=L(420,400,a),w=400,h=532,m=E((p-.22-i*.1)/.4);

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glass(cx0,y,w,h,24,a);halo(cx0+110,y+120,160,K[0],.45*m*a);
T(o.co,cx0+34,y+56,22,KT[3],'left',700,a);T(Math.round(o.v*m)+'%',cx0+30,y+158,110,K[0],'left',
700,a,DF);
X.font=`700 26px ${DF}`;const sgl=wrap(o.sg,cx0+34,y+206,w-68,30,26,INK,700,a,DF);
const
ly=wrap(o.src,cx0+34,sgl+44,w-68,28,21,INK,600,a);wrap(o.d,cx0+34,ly+40,w-68,28,21,MUT,50
0,a);
const
bx=cx0+34,bw=w-68,byy=y+h-44;X.globalAlpha=a;X.fillStyle=LN;X.beginPath();X.roundRect(bx,
byy,bw,10,5);X.fill();
const
gr=X.createLinearGradient(bx,0,bx+bw,0);gr.addColorStop(0,K[2]);gr.addColorStop(1,K[0]);X.fill
Style=gr;X.beginPath();X.roundRect(bx,byy,Math.max(10,bw*o.v/100*m),10,5);X.fill();X.globalAl
pha=1;
T('Benchmark similarity',bx,byy-14,20,MUT,'left',600,a)}}
const RUNS=[];for(let i=0;i<6;i++)for(let
j=0;j<11;j++)RUNS.push({i,j,off:(R()-5)*60,far:(R()-5)*640,ap:.08+(j*6+i)/66*.5});
function S_conf(p,t){T('Same study, re-run with virtual personas',130,380,30,INK,'left',700,1,DF);
const
s=E((p-.2)/.6),bw=L(560,90,s);X.fillStyle=rgba(K[3],.09);X.fillRect(665-bw/2,420,bw,420);Ln(665,
410,665,850,K[3],2.5,.9,[8,8]);T('Target result',665,882,21,KT[3],'center',600);
for(let i=0;i<6;i++){const y=450+i*70;T('Context
'+(i+1),130,y+7,21,MUT,'left',600);Ln(280,y,1050,y,LN,2)}
RUNS.forEach(r=>{const a=E((p-r.ap)/.05);if(a<=0)return;const
q=EO((p-r.ap)/.35);D(665+L(r.far,r.off,q),450+r.i*70,8,K[3],a*.85));
const cx=1470,cy=770,R2=250,g=.12+.8*E((p-.4)/.45),ang=Math.PI+Math.PI*g;
T('Virtual World Confidence',cx,420,32,INK,'center',700,1,DF);halo(cx,cy-80,320,K[3],.4*g);
X.lineCap='round';X.lineWidth=30;X.strokeStyle=LN;X.beginPath();X.arc(cx,cy,R2,Math.PI,TAU);
X.stroke();
const
gr=X.createLinearGradient(cx-R2,0,cx+R2,0);gr.addColorStop(0,K[1]);gr.addColorStop(1,K[3]);X
.strokeStyle=gr;X.beginPath();X.arc(cx,cy,R2,Math.PI,ang);X.stroke();X.lineCap='butt';
Ln(cx,cy,cx+Math.cos(ang)*(R2-50),cy+Math.sin(ang)*(R2-50),INK,5);D(cx,cy,14,INK);
T('Low',cx-R2,cy+50,20,MUT,'center',600);T('High',cx+R2,cy+50,20,MUT,'center',600);
const hi=g>.8;T(hi?'High confidence':'Building
confidence',cx,cy-110,40,hi?KT[3]:INK,'center',700,1,DF);T('Results hold across repeated
runs',cx,cy+100,22,MUT,'center',600,E((p-.7)/.2))}

function S_intro(p,t){const cx=500,cy=520,a=EO(p/.3);
halo(cx,cy,380,K[3],.6*a);halo(cx+120,cy-160,220,K[1],.5*a);
X.save();X.globalAlpha=a;X.lineWidth=5;X.strokeStyle='#EC6A9E';X.beginPath();X.arc(cx,cy,30
0,0,TAU*EO(p/.45));X.stroke();

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```

X.strokeStyle='#5FCDEA';X.beginPath();X.arc(cx,cy,185,Math.PI,Math.PI+TAU*EO((p-.08)/.45));
X.stroke();X.restore();
D(cx,cy,82*EO((p-.15)/.3),'#5FCDEA',1);
[[300,-.9,11,'#E0457F',.00032],[300,2.9,10,'#F29BBE',.00032],[300,1.1,9,'#F29BBE',.00032],[18
5,-.2,9,'#5FCDEA',-.0005],[185,2.4,6,'#F29BBE',-.0005],[185,-2.3,6,WH,-.0005]].forEach((o,i)=>{
const
an=o[1]+t*o[4],b=E((p-.35-i*.05)/.2);D(cx+Math.cos(an)*o[0],cy+Math.sin(an)*o[0],o[2],o[3],b);if(o
[3]==WH){X.globalAlpha=b;X.strokeStyle='#EC6A9E';X.lineWidth=2;X.beginPath();X.arc(cx+M
ath.cos(an)*o[0],cy+Math.sin(an)*o[0],o[2],0,TAU);X.stroke();X.globalAlpha=1}});
const b=E((p-.25)/.3);Ln(900,300,900,300+440*b,INK,4,b);
T('The Human',960,420,104,INK,'left',700,E((p-.3)/.25),DF);T('Insights
Engine',960,530,104,INK,'left',700,E((p-.36)/.25),DF);
T('Real world evidence, simulated feedback, informed
decisions.',964,620,26,KT[0],'left',700,E((p-.5)/.25));
wrap('The Human Insights Engine is an evidence-informed and AI-powered simulation tool that
helps development organizations explore potential responses to a proposed policy, program,
product, service, or message before resources are committed. It turns relevant real-world
evidence into directional hypotheses for a defined population, geography, behavioral question,
and use case.',964,700,820,38,24,MUT,500,E((p-.62)/.25))}

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```

const SLIDES=[
{intro:1,title:"",sub:"",steps:['Overview'],fn:[S_intro],dur:[9000],cap:['']},
{title:'From a shell to dynamic virtual worlds',sub:'How the Human Insights Engine is built, one
layer at a time',
steps:['Shell','Train','Segments','Calibrate','Virtual
personas','Interact','Worlds'],fn:[S_shell,S_train,S_seg,S_cal,S_people,S_int,S_worlds],dur:[700
0,7500,6500,7500,7000,8000,9500],
cap:['Every virtual persona starts as a shell: demographics, psychographics, behaviours and
context, each defined by its own variables.',
'The shell is trained on 3.2 billion real-world research data points, from evidence representing
1.6 million real-world participants. Inputs only.',
'The trained shell generates virtual persona segments for each region: static profiles you can
interact with.',
'Accuracy improves only by feeding in richer, higher-quality input data, never outputs. As inputs
improve, weights settle and the shell becomes dynamic.',
'Slight variations on each variable turn one segment into hundreds of distinct virtual personas.',
'Virtual personas interact, form groups, and engage with money, institutions, services and
products.',
'Together they form dynamic virtual worlds, generated on the go: 59.5 billion simulation data
points so far.'],
{title:'Tested against real-world human studies',sub:'Four backward-validation exercises across
agriculture, financial inclusion and health',
steps:['Accuracy','Confidence'],fn:[S_acc,S_conf],dur:[9000,9000],

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cap:['HIE virtual personas see only a study\u2019s inputs. Their responses matched real-world results with 94% to 99.5% benchmark similarity.',
'Results hold over multiple runs of the same study with HIE virtual personas, at a substantially lower sample size.']]

```
const LOGO=`<svg viewBox="0 0 100 100" aria-hidden="true"><circle cx="50" cy="50" r="44" fill="none" stroke="#EC6A9E" stroke-width="2.4"/><circle cx="50" cy="50" r="27" fill="none" stroke="#5FCDEA" stroke-width="2.4"/><circle cx="50" cy="50" r="12" fill="#5FCDEA"/><circle cx="82" cy="20" r="5.5" fill="#E0457F"/><circle cx="8" cy="42" r="5" fill="#F29BBE"/><circle cx="74" cy="41" r="4.6" fill="#5FCDEA"/><circle cx="33" cy="30" r="3" fill="fff" stroke="#EC6A9E" stroke-width="1"/><circle cx="36" cy="72" r="3.2" fill="#F29BBE"/><circle cx="76" cy="87" r="5" fill="#F29BBE"/></svg>`;
const IC={prev:<svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2" stroke-linecap="round" stroke-linejoin="round"><path d="M15 18l-6-6 6-6"/></svg>,next:<svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2" stroke-linecap="round" stroke-linejoin="round"><path d="M9 18l6-6 6-6"/></svg>,
pause:<svg viewBox="0 0 24 24" fill="currentColor"><rect x="6" y="5" width="4" height="14" rx="1"/><rect x="14" y="5" width="4" height="14" rx="1"/></svg>,play:<svg viewBox="0 0 24 24" fill="currentColor"><path d="M7 5v14l12-7z"/></svg>,
fs:<svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2" stroke-linecap="round" stroke-linejoin="round"><path d="M4 9V4h5M20 9V4h-5M4 15v5h5M20 15v5h-5"/></svg>};
const reduce=matchMedia('(prefers-reduced-motion: reduce)').matches;
function mount(fig,S){
fig.innerHTML=`<div class="frame"><div class="stage"><canvas
aria-hidden="true"></canvas><header><h5></h5><p class="sub"></p></header><div
class="logo">${LOGO}</div><span>The Human<br>Insights Engine</span></div><nav
class="rail" aria-label="Animation steps"></nav></div></div>
<div class="hbar"><button class="pv" aria-label="Previous step">${IC.prev}</button><button
class="play" aria-label="Pause"><span class="ico">${IC.pause}</span><span
class="lb">Pause</span></button><button class="nx" aria-label="Next
step">${IC.next}</button><p class="cap" aria-live="polite"><p><button class="fsb"
aria-label="Full screen">${IC.fs}</button></p></div>`;
const
q=c=>fig.querySelector(c),frame=q('.frame'),stg=q('.stage'),cv=q('.canvas'),X0=cv.getContext('2d'),off=document.createElement('canvas'),XO=off.getContext('2d');
q('h5').textContent=S.title;q('.sub').textContent=S.sub;
let sc=1,st=0,start=performance.now(),playing=!reduce,vis=false,seen=false;
const btns=S.steps.map((n,i)=>{const
b=document.createElement('button');b.innerHTML=`<span class="bar"><b></b></span><span
class="n">${String(i+1).padStart(2,'0')}</span>${n}`;b.setAttribute('aria-label','Step ${i+1}: ${n}')b.onclick=()=>go(i);q('.rail').appendChild(b);return b});
function fit(){const w=frame.clientWidth,h=frame.clientHeight;if(!w||!h)return;const
s=Math.min(w/1920,h/1080);stg.style.transform=`translate(${(w-1920*s)/2}px,${(h-1080*s)/2}px)`
```

```

scale(${s})`;const
d=Math.min(2,window.devicePixelRatio||1);sc=s*d;cv.width=off.width=Math.max(1,Math.round(1
920*sc));cv.height=off.height=Math.max(1,Math.round(1080*sc))}
new ResizeObserver(fit).observe(frame);fit();
function
go(k){st=(k+S.steps.length)%S.steps.length;start=performance.now();btns.forEach((b,i)=>{b.clas
sList.toggle('on',i===st);b.setAttribute('aria-current',i===st?'step':'false')});q('.cap').textContent=S.
cap[st]}
function setPlay(v){playing=v;q('.play .ico').innerHTML=v?!C.pause:IC.play;q('.play
.lb').textContent=v?'Pause':'Play';q('.play').setAttribute('aria-label',v?'Pause':'Play')}
q('.pv').onclick=()=>go(st-1);q('.nx').onclick=()=>go(st+1);q('.play').onclick=()=>setPlay(!playing);
q('.fsb').onclick=()=>{try{document.fullscreenElement?document.exitFullscreen():fig.requestFulls
creen()}catch(e){}};
fig.addEventListener('keydown',e=>{if(e.target.closest('button')&&(e.key==='
'|e.key==='Enter'))return;
if(e.key===' '){e.preventDefault();setPlay(!playing)}else
if(e.key==='ArrowRight'){e.preventDefault();go(st+1)}else
if(e.key==='ArrowLeft'){e.preventDefault();go(st-1)}else
if(e.key==='f'||e.key==='F')q('.fsb').click()});
new
IntersectionObserver(es=>es.forEach(en=>{vis=en.isIntersecting;if(vis&&!seen){seen=true;go(0)
}},{threshold:.35}).observe(frame);
setPlay(playing);go(0);
function loop(now){if(vis&&cv.width>1){const
e=now-start,dur=S.dur[st],p=Math.min(1,Math.max(0,e/(dur*.72)));
if(playing&&e>dur){go(st+1);return requestAnimationFrame(loop)}
X=XO;SC=sc;XO.setTransform(1,0,0,1,0,0);XO.clearRect(0,0,off.width,off.height);XO.setTransfo
rm(sc,0,0,sc,0,0);S.fn[st](p,now);
XO.setTransform(1,0,0,1,0,0);XO.clearRect(0,0,cv.width,cv.height);XO.globalAlpha=Math.min(1,e/
450);XO.drawImage(off,0,0);XO.globalAlpha=1;
btns.forEach((b,i)=>{b.firstChild.firstChild.style.width=(i<st?100:i>st?0:(playing?Math.min(1,e/du
r):p)*100)+'%'}}}
requestAnimationFrame(loop)}
requestAnimationFrame(loop)}
const
go2=()=>document.querySelectorAll('figure.hie').forEach(f=>mount(f,SLIDES[+f.dataset.hie]));
(document.fonts&&document.fonts.ready?document.fonts.ready:Promise.resolve()).then(go2,go
2);
})();
</script>
</body>
</html>

```